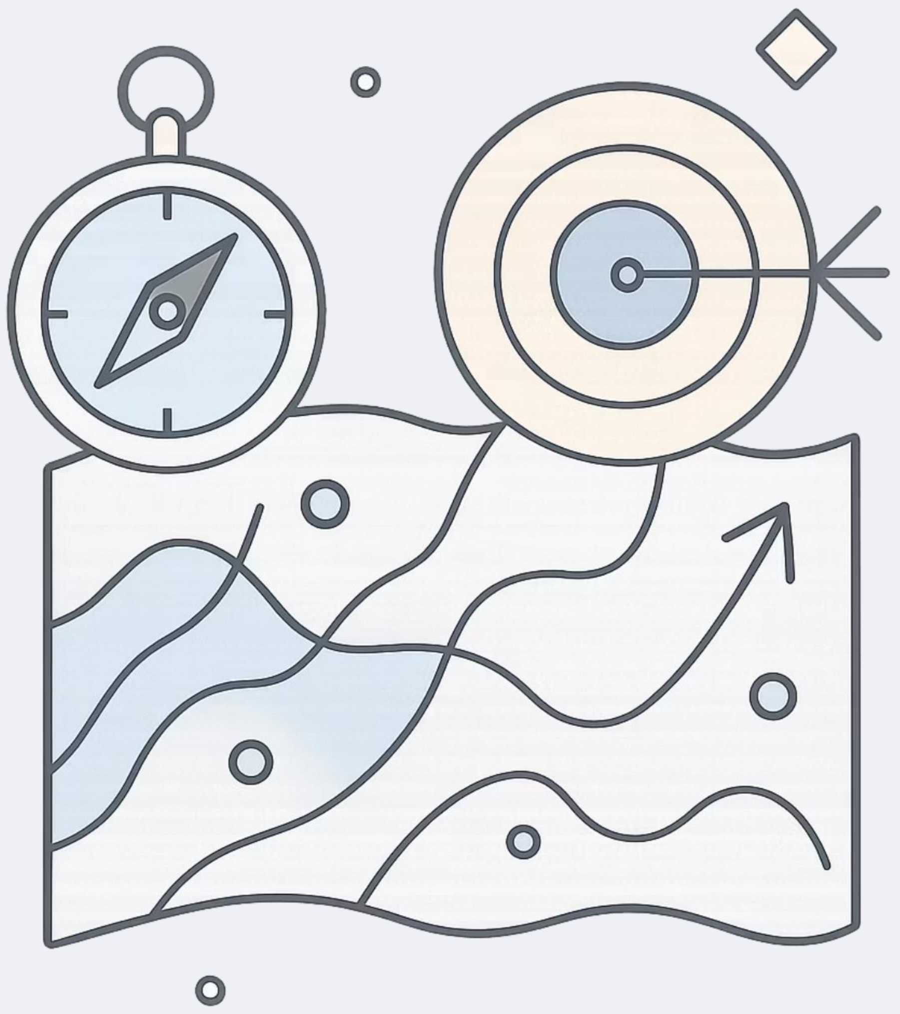
Scheduling

Mastering Time and Resources

IMG 401

SCHEDULING FUNDAMENTALS



Why Schedule?

Scheduling is the process of defining activities, sequencing them, estimating resources and durations, and developing a plan to guide execution and control.

🎯 Achieve Objectives

Align work to goals with clarity and purpose

⚙️ Manage Resources

Use people, equipment, and materials efficiently

📅 Ensure Timeliness

Drive on-time delivery across all industries

The Core Components of Scheduling

Every effective schedule is built on five interconnected pillars. Together, they form the backbone of any successful plan — whether you're managing a hospital, a factory, or a product launch.

1

Activities

Breaking down work into manageable, clearly defined tasks

2

Sequencing

Determining the logical order in which activities must be performed

3

Resource Estimation

Identifying and allocating personnel, equipment, and materials

4

Duration Estimation

Predicting the time required for each individual activity

5

Schedule Development

Synthesizing all inputs into a cohesive, executable plan

Visualizing the Flow: Understanding Dependencies

Dependencies define the **logical relationships** between activities. Choosing the right type ensures your schedule reflects reality accurately.

1

Finish-to-Start (FS)

B cannot *start* until A *finishes* — the most common relationship

2

Start-to-Start (SS)

B cannot *start* until A *starts* — useful for parallel work

3

Finish-to-Finish (FF)

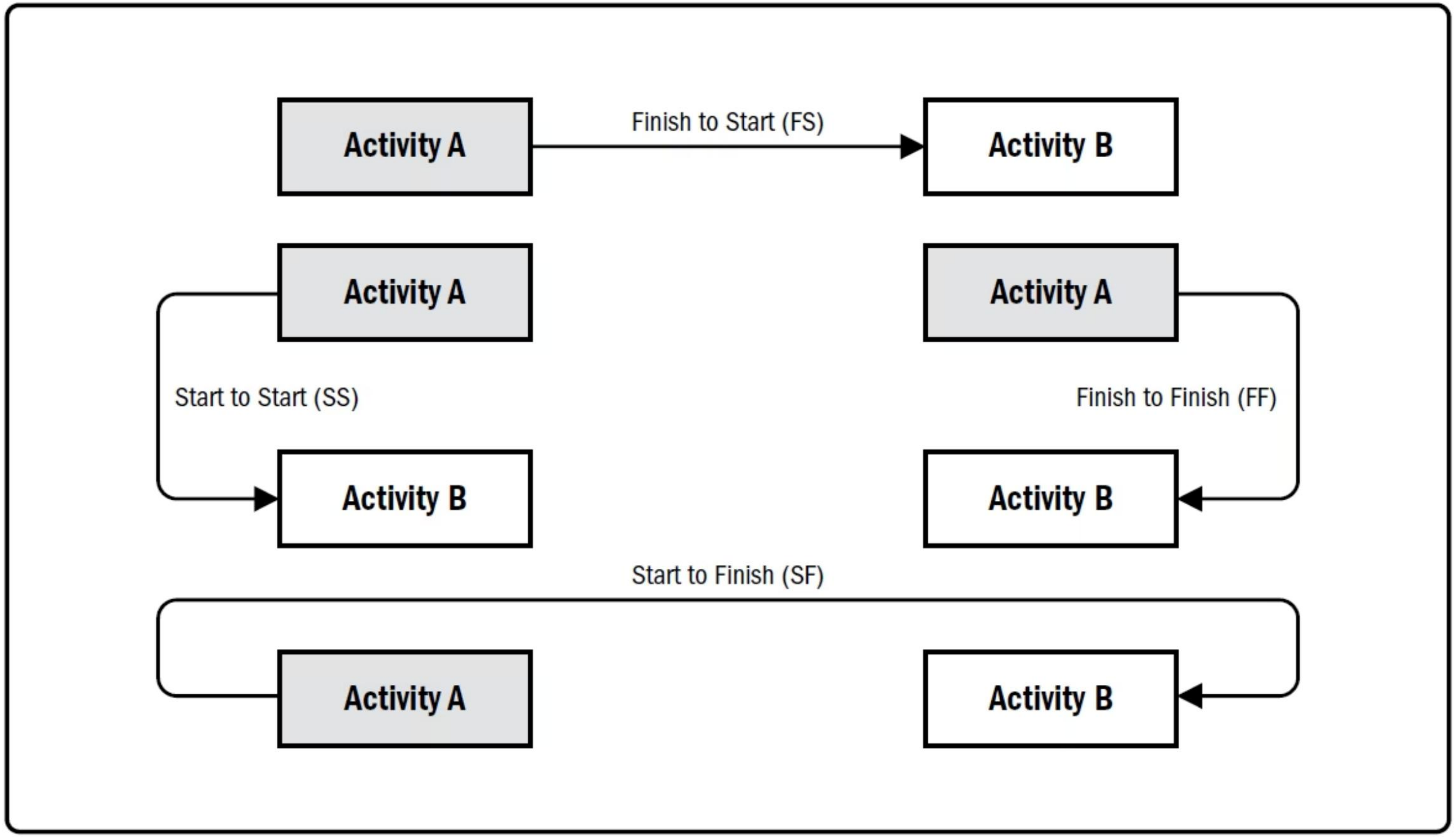
B cannot *finish* until A *finishes* — linked completion

4

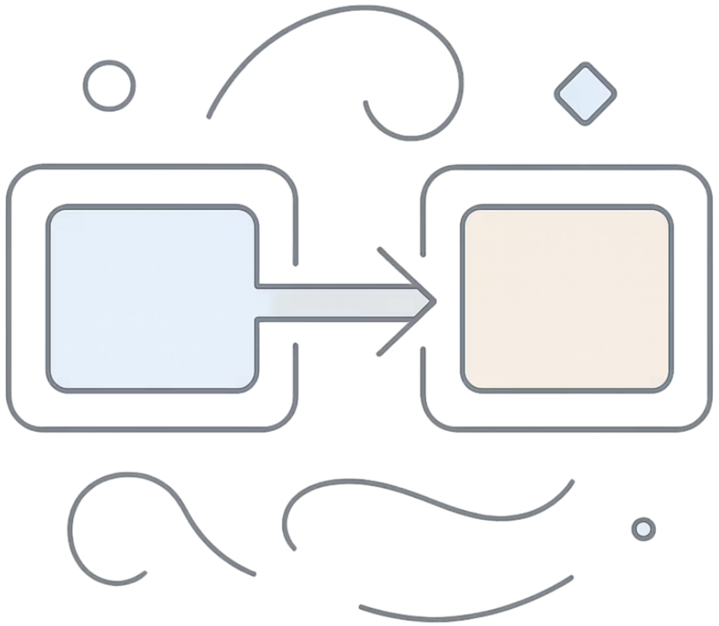
Start-to-Finish (SF)

B cannot *finish* until A *starts* — rare, used in shift handovers

Precedence Diagram



Precedence Diagramming Method (PDM) Relationship Types



KEY CONCEPT

Finish-to-Start in Practice

The **Finish-to-Start** dependency is the most intuitive: you cannot begin laying a roof until the walls are standing, and you cannot ship a product before it's assembled.

Understanding dependencies prevents costly overlaps and ensures teams don't begin work before prerequisites are complete.

Task B waits for Task A — simple, but foundational to every schedule.

Key Scheduling Techniques

Two dominant methods help professionals structure and analyze schedules — each suited to different levels of complexity and uncertainty.

📊 Gantt Charts

Visual timelines displaying activity durations, start/end dates, and dependencies. Ideal for communicating progress to stakeholders and tracking work at a glance.

🔗 Network Diagrams (PERT/CPM)

Illustrate the logical flow of activities and identify the **critical path** — the sequence of tasks that directly controls when the work will be complete.

Gantt Chart

A well-structured Gantt chart is one of the most universally understood scheduling tools — used in construction, healthcare, software, events, and beyond.

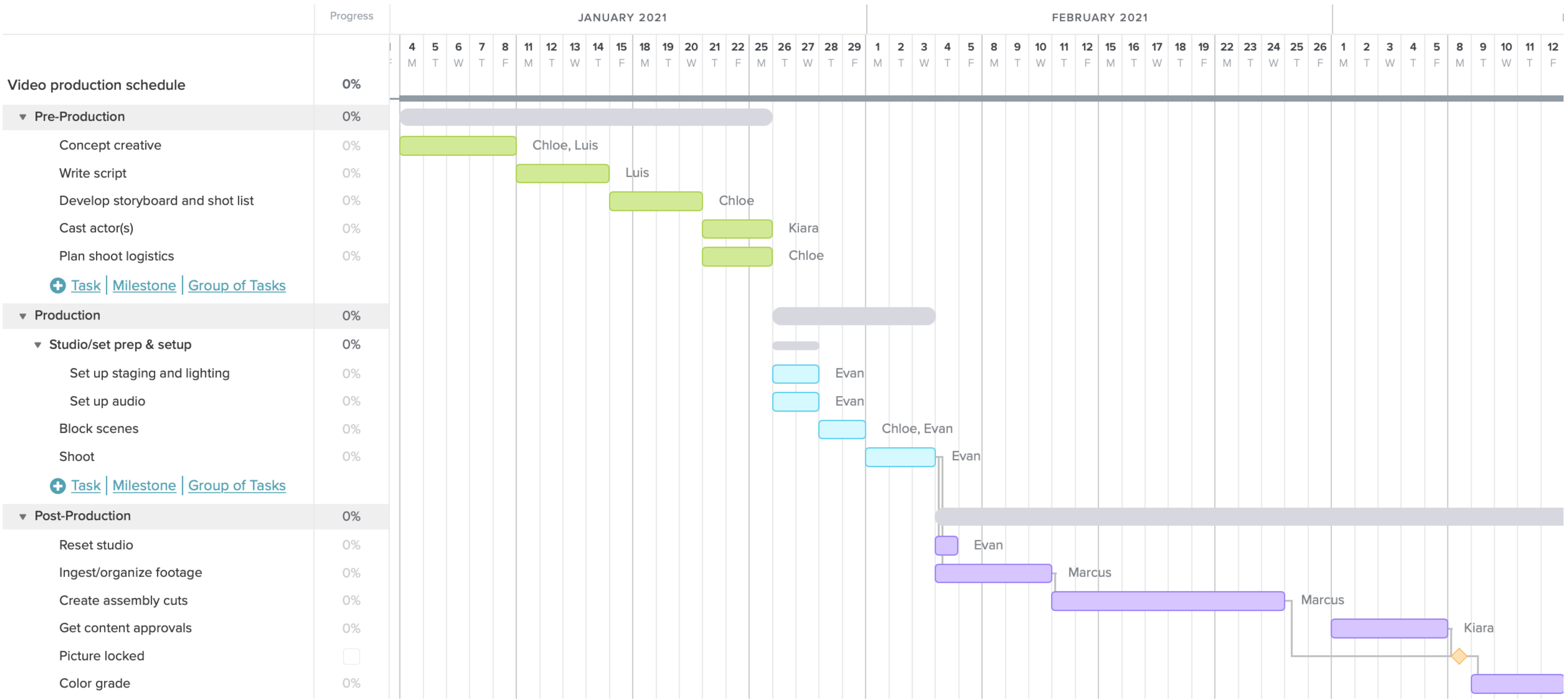


Figure: Gantt Chart

Schedule Network Diagram

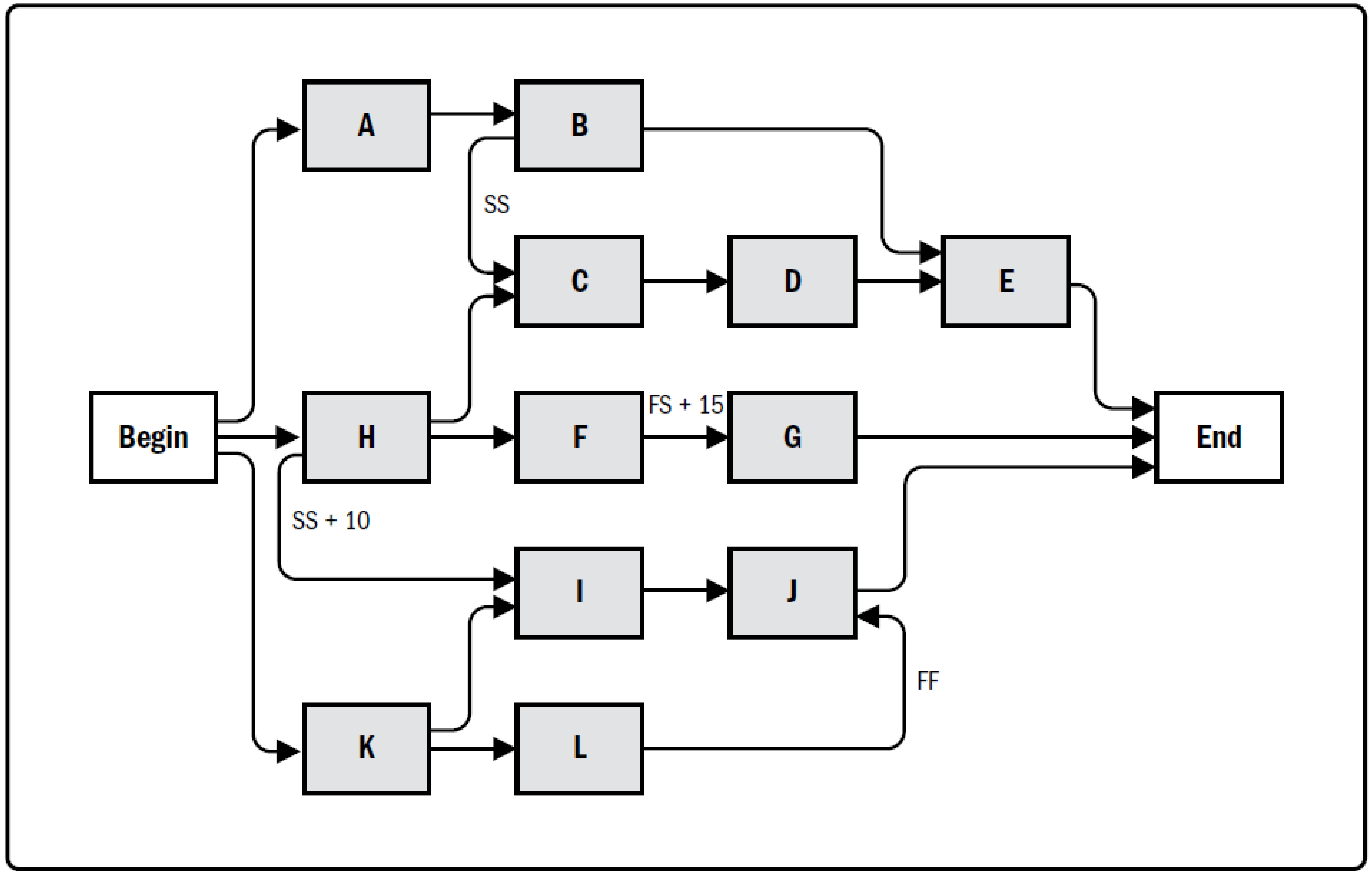
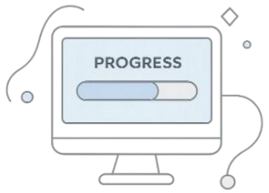


Figure: . Project Schedule Network Diagram

Gantt Chart Spotlight: Seeing the Big Picture

Visual Timeline

Clearly shows start and end dates for every activity at a single glance

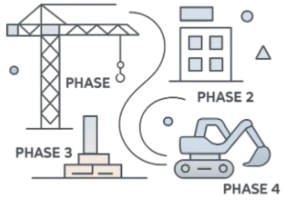


Progress Tracking

Completed work vs. remaining tasks visualized as filled bars — instantly readable

Resource Allocation

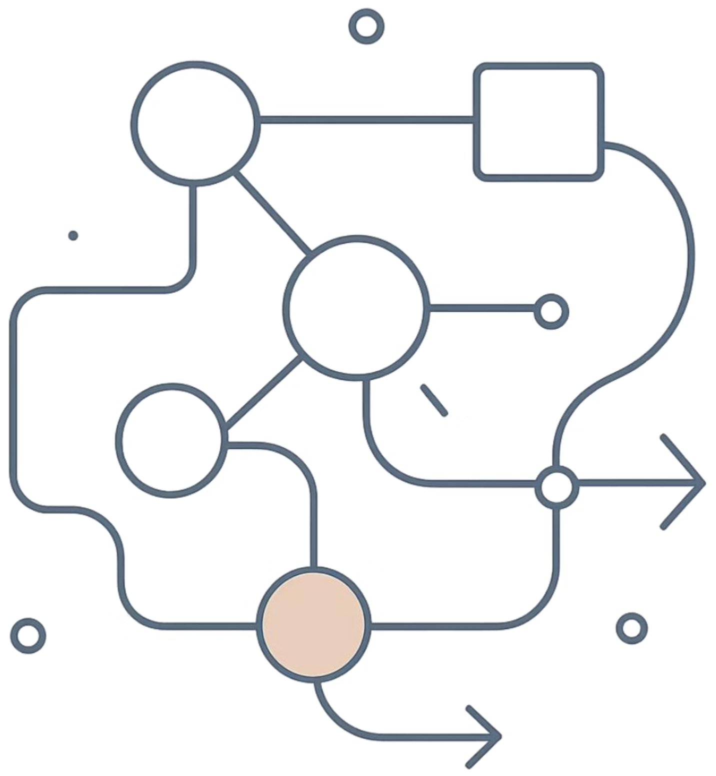
Assigned owners and resources displayed per task to prevent conflicts



Real-World Example

Construction: Foundation → Framing → Roofing → Interior, each phase sequenced and tracked

Critical Path Method (CPM): The Race Against Time



What is the Critical Path?

The **longest sequence** of dependent activities that determines the **earliest possible completion** of the entire work effort.

Any delay = Total delay

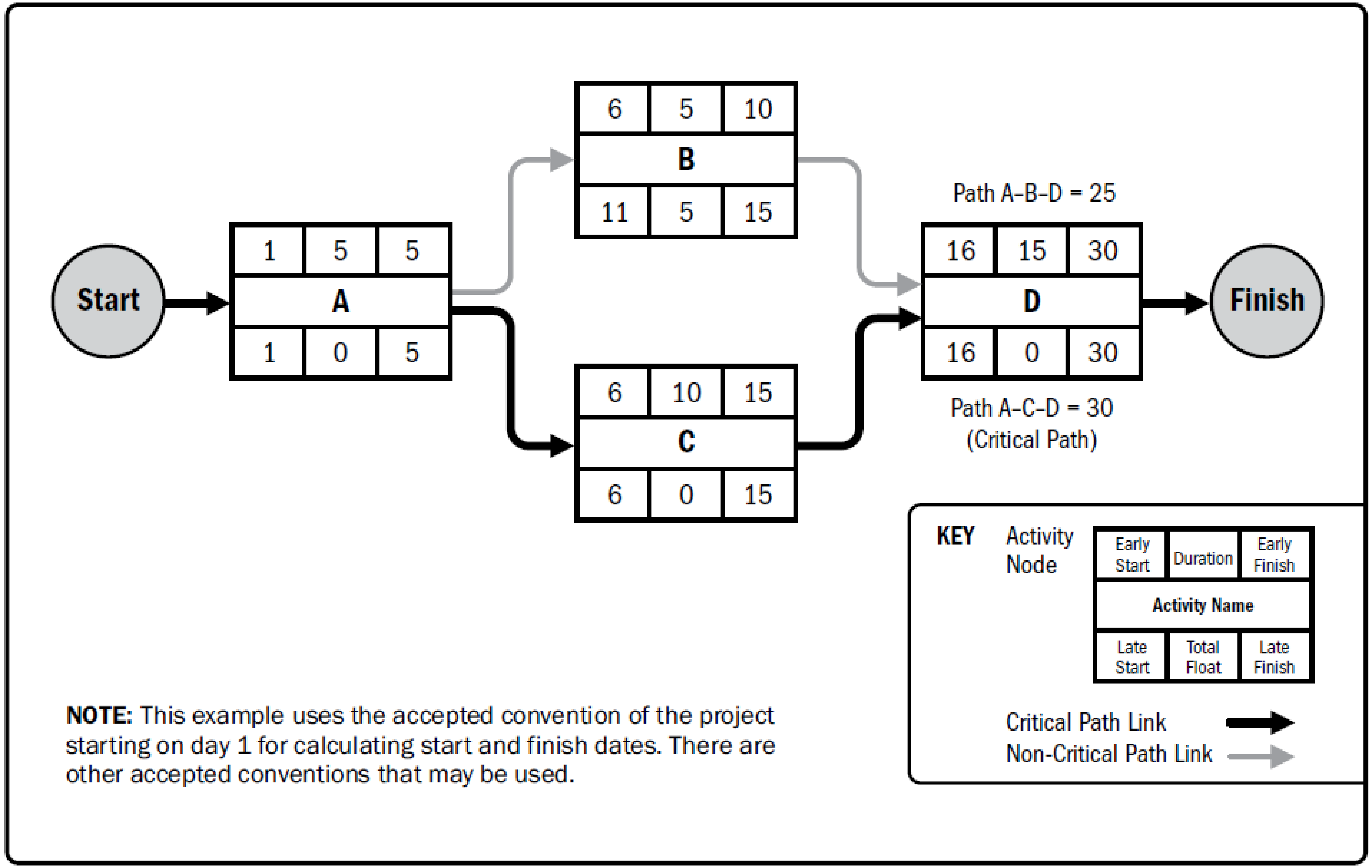
A slip on any critical task directly pushes the final deadline

Focus management attention

CPM reveals where to concentrate resources and monitoring effort

Float on non-critical tasks

Tasks off the critical path have flexibility — use it strategically



Example of Critical Path Method

Critical Path Example: Manufacturing Assembly Line

Identifying the critical path exposes which activities hold the entire production process hostage — and which have room to flex.

If **Sub-assembly 1** is delayed — due to a supplier issue, machine failure, or staffing gap — every downstream step shifts, and the product launch date moves with it.

PERT: Handling Uncertainty in Duration Estimates

Three-Point Estimation

PERT acknowledges that real-world durations are rarely certain. By combining three scenarios, it produces a statistically grounded expected duration.

- **Triangular distribution.** $cE = (cO + cM + cP) / 3$
- **Beta distribution.** $cE = (cO + 4cM + cP) / 6$

PERT Formula
Expected Duration = (O + 4M + P) ÷ 6
The most likely estimate is weighted **4×** more heavily, reflecting its higher probability.

This approach is especially valuable in **R&D, software development, and healthcare** — any environment where uncertainty is inherent and single-point estimates are misleading.

Optimistic (O)

Best-case scenario if everything goes right

Most Likely (M)

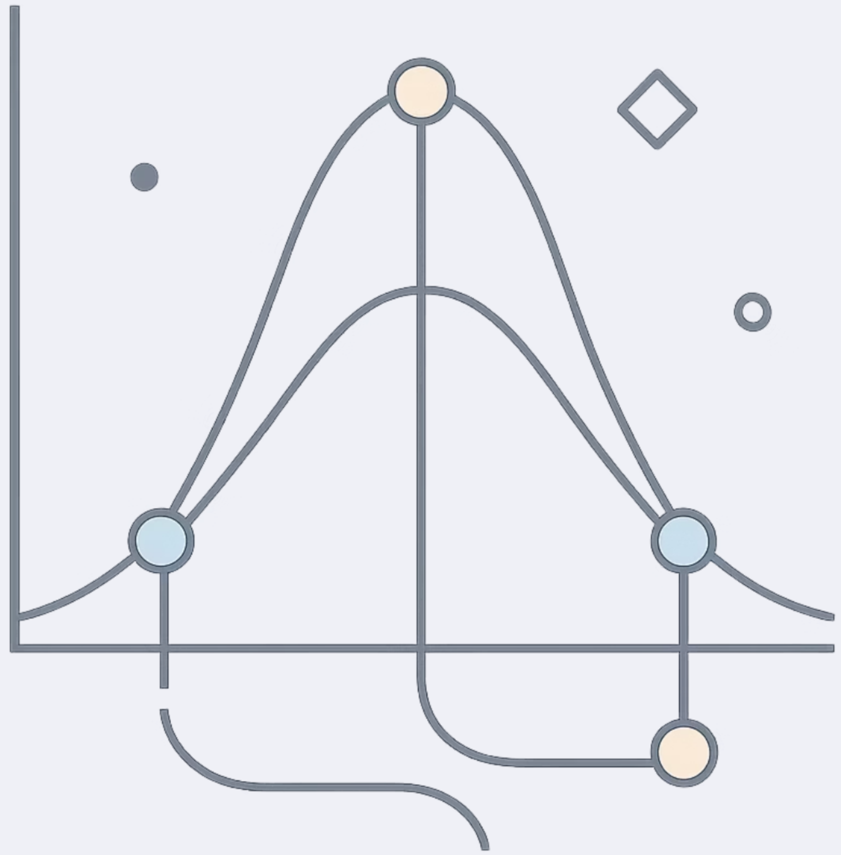
The most realistic, probable duration

Pessimistic (P)

Worst-case if major problems arise

Visualizing PERT: The Probability Distribution

The three-point PERT estimate forms a **weighted distribution**. The peak represents the most likely outcome, while the tails capture optimistic and pessimistic scenarios — giving planners a realistic range, not just a single number.



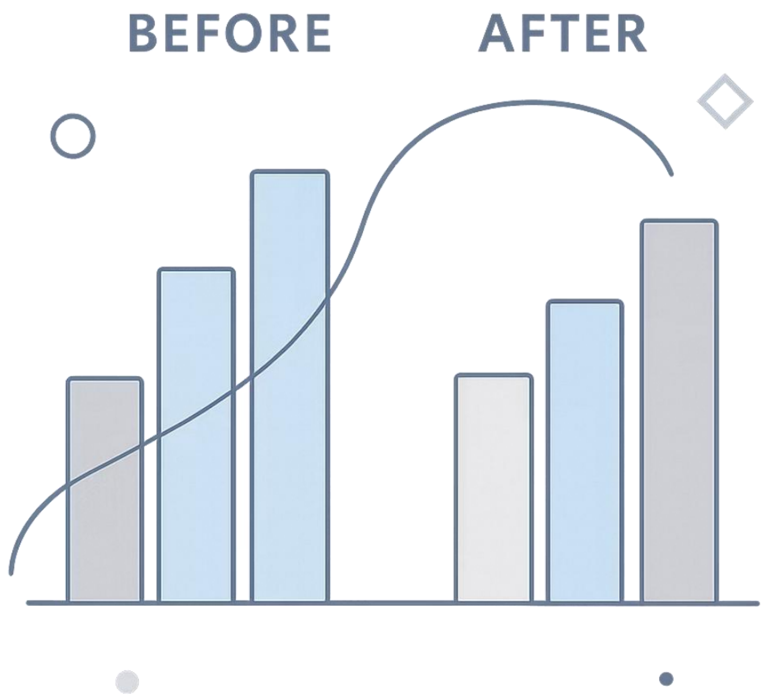
Resource Leveling & Smoothing: Optimizing Your Assets

⚖️ Resource Leveling

Adjusts activity start and end dates to **resolve over-allocation conflicts**. Ensures no resource is stretched beyond capacity. *May extend the overall schedule.*

🔧 Resource Smoothing

Adjusts activities within their available **float** to optimize usage — without extending the schedule. A softer approach when the deadline is fixed.



Rule of thumb: Use **leveling** when resource conflicts are severe. Use **smoothing** when the deadline cannot move.

Resource Optimization

Resource optimization is used to adjust the start and finish dates of activities to adjust planned resource use to be equal to or less than resource availability. Examples of resource optimization techniques that can be used to adjust the schedule model due to demand and supply of resources include but are not limited to:

Resource leveling: A technique in which start and finish dates are adjusted based on resource constraints with the goal of balancing the demand for resources with the available supply. Resource leveling can be used when shared or critically required resources are available only at certain times or in limited quantities, or are overallocated, such as when a resource has been assigned to two or more activities during the same time period, or there is a need to keep resource usage at a constant level. Resource leveling can often cause the original critical path to change. Available float is used for leveling resources. **Consequently, the critical path through the project schedule may change.**

Resource smoothing: A technique that adjusts the activities of a schedule model such that the requirements for resources on the project do not exceed certain predefined resource limits. In resource smoothing, as opposed to resource leveling, **the project's critical path is not changed and the completion date may not be delayed**. In other words, activities may only be delayed within their free and total float. **Resource smoothing may not be able to optimize all resources.**

Schedule Compression

Schedule compression techniques are used to shorten or accelerate the schedule duration without reducing the project scope in order to meet schedule constraints, imposed dates, or other schedule objectives. A helpful technique is the negative float analysis. The critical path is the one with the least float. Due to violating a constraint or imposed date, the total float can become negative.

Crashing: A technique used to shorten the schedule duration for the least incremental cost by adding resources. Examples of crashing include approving overtime, bringing in additional resources, or paying to expedite delivery to activities on the critical path. **Crashing works only for activities on the critical path where additional resources will shorten the activity's duration.** Crashing does not always produce a viable alternative and may result in increased risk and/or cost.

Fast tracking: A schedule compression technique in which activities or phases normally done in sequence are performed in parallel for at least a portion of their duration. An example is constructing the foundation for a building before completing all of the architectural drawings. Fast tracking may result in rework and increased risk. **Fast tracking only works when activities can be overlapped to shorten the project duration on the critical path.** Fast tracking may also increase project costs.

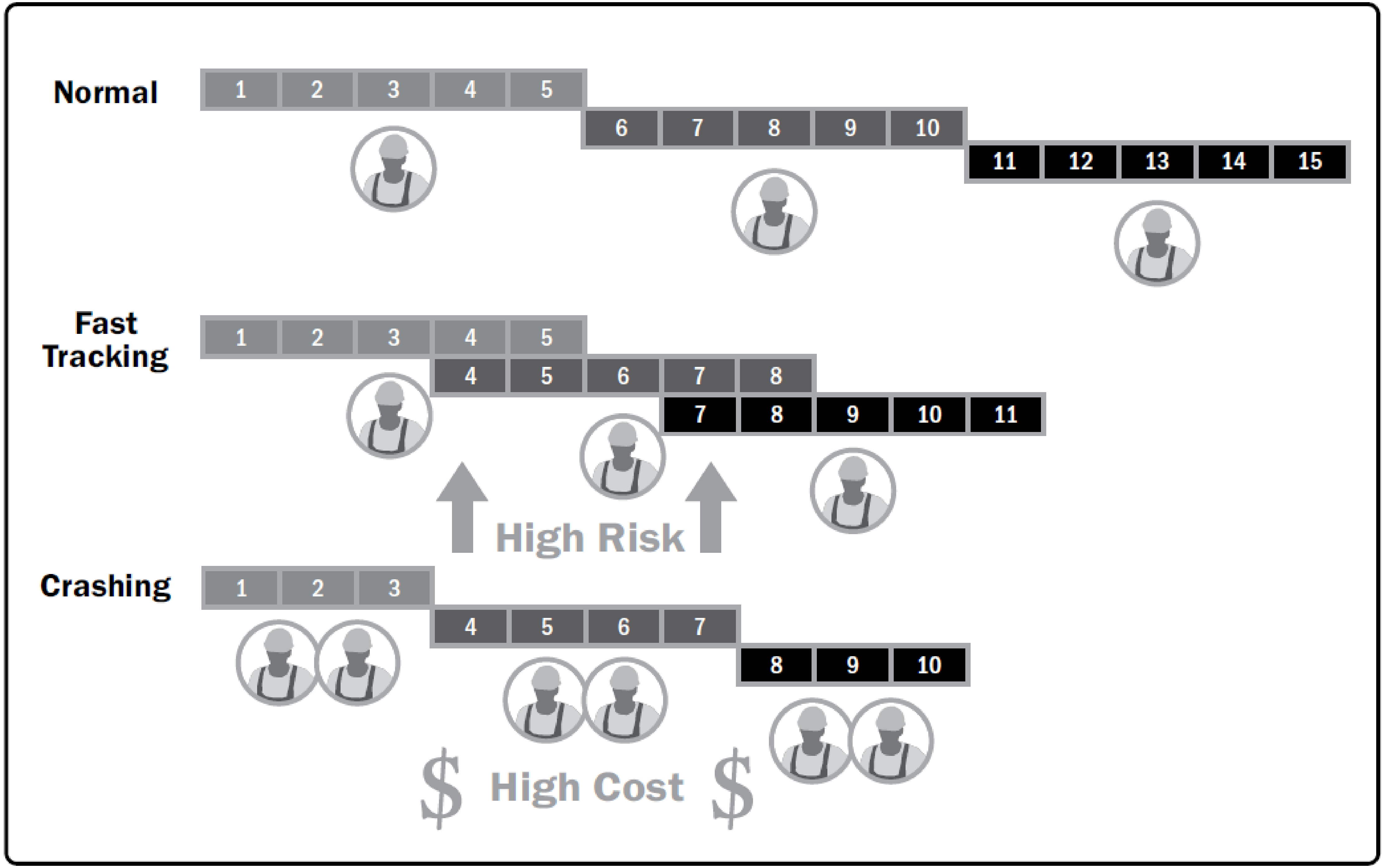


Figure: Schedule Compression Comparison

Scheduling in Action: Across Every Industry

Scheduling principles are universal — the tools and rhythms vary by industry, but the core logic of sequencing, dependencies, and resource management applies everywhere.

Healthcare

Patient appointment scheduling, OR management, and nurse rostering to maximize care capacity

Manufacturing

Production line sequencing, inventory replenishment, and supply chain logistics coordination

Software Development

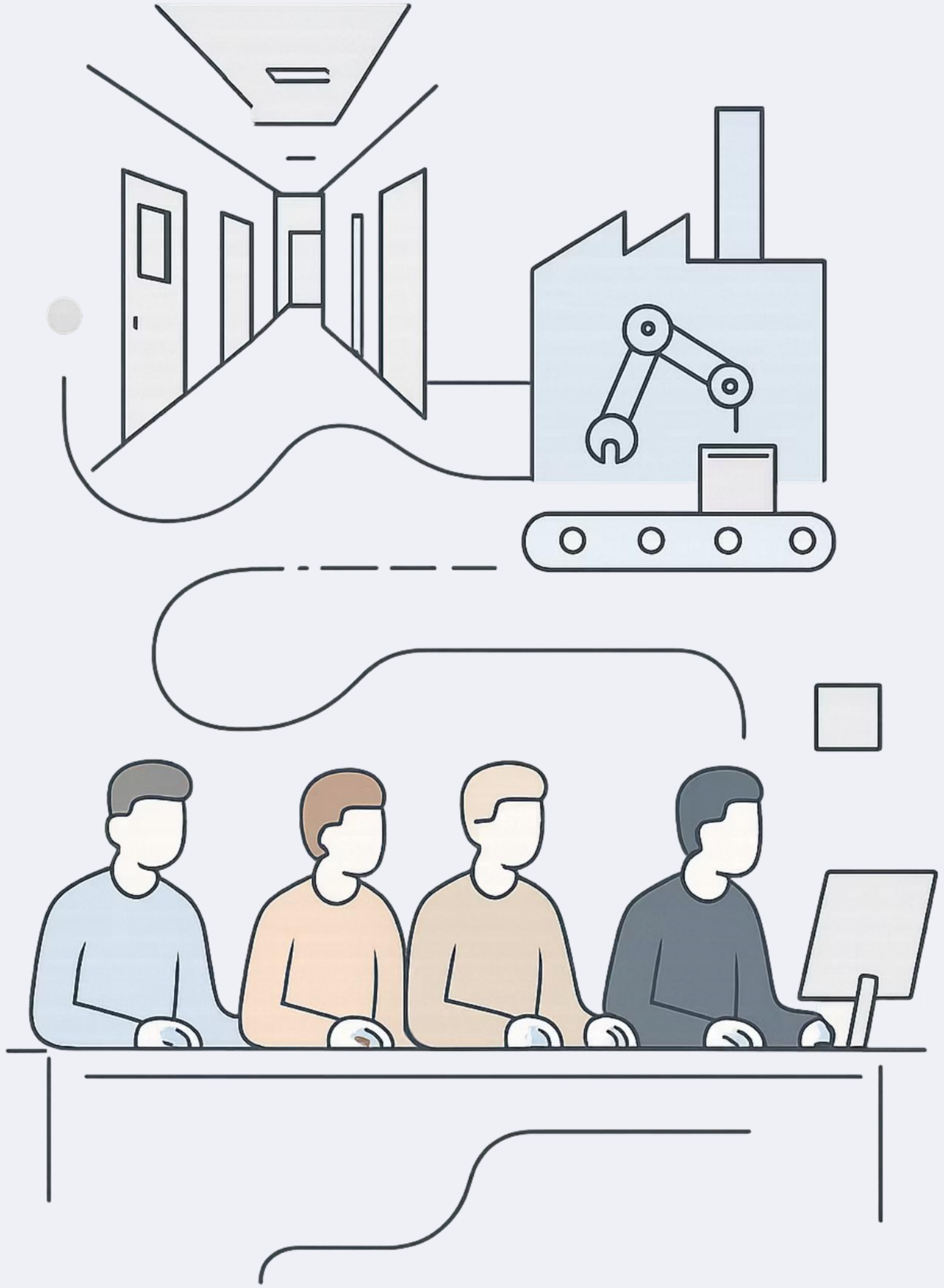
Agile sprints, release planning, and feature development timelines tied to user stories

Event Management

Conference planning, venue booking, vendor coordination, and speaker scheduling

One Discipline, Many Arenas

From a surgeon's operating schedule to a manufacturer's shift roster, scheduling shapes outcomes in every professional context. The industry changes — the principles don't.



Challenges in Scheduling

Even the best-crafted schedule will face disruption. Recognizing common failure modes is the first step to building resilience.

Unforeseen Events

Weather delays, equipment failures, illness, or supply shortages can derail even well-buffered plans

Scope Creep

Uncontrolled additions to requirements silently consume time and resources — without adjusting the schedule

Inaccurate Estimates

Overly optimistic durations or underestimated resources are among the most common root causes of schedule failure

Communication Breakdowns

When teams aren't aligned on status, changes, or blockers, the schedule drifts silently until it's too late

Best Practices for Effective Scheduling

01

Define Clear Objectives

Anchor the schedule to measurable outcomes, not just activities

02

Involve Stakeholders Early

Gather input from those who will execute and be affected by the schedule

03

Use Appropriate Tools

Match software complexity to the scale of your work

04

Monitor and Update Regularly

Schedules are living documents — revisit them as conditions change

05

Communicate Proactively

Surface delays and changes early; surprises cost more the later they arrive

The Iterative Scheduling Loop

Scheduling is not a one-time event. Successful practitioners treat it as a **continuous cycle** — revisiting and refining as real-world data replaces assumptions.

Plan

Define activities, sequences, and resource needs

Execute

Carry out work according to the schedule

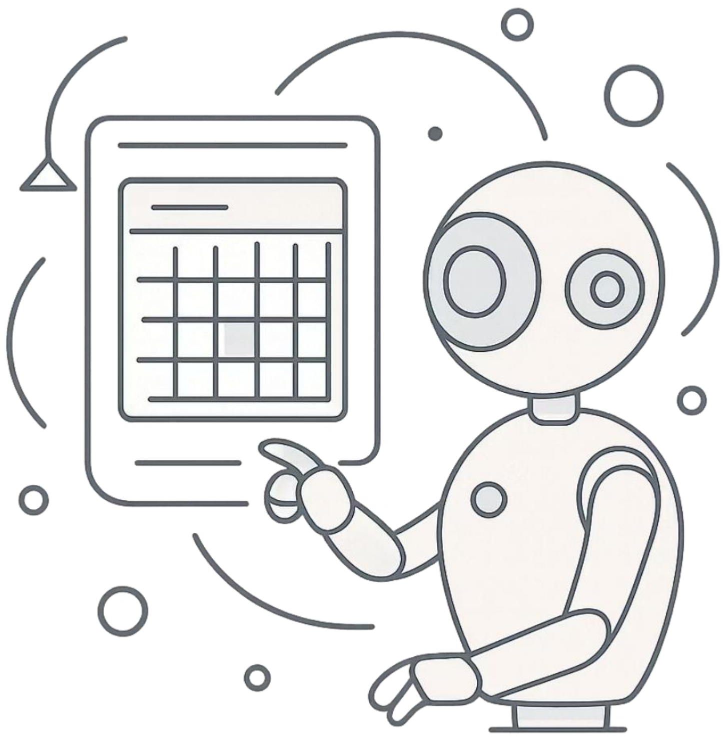
Control

Take corrective action and update the schedule accordingly

Monitor

Track actual progress against the baseline plan

The Future of Scheduling: Automation & AI



What's Changing?

Emerging technologies are transforming scheduling from a manual discipline into an **intelligent, adaptive system**.

Smart Scheduling Software

Automates complex calculations, resource conflict resolution, and real-time rescheduling

AI-Powered Predictive Analytics

Flags risks before they become delays by learning from historical patterns

Cross-Platform Integration

Connects calendars, ERP systems, and team tools into one unified scheduling layer

Mastering Your Schedule: The Key to Achievement

Effective scheduling is not simply about creating a plan — it is about **proactive management, clear communication, and disciplined adaptability**.

Understand the Tools

Gantt, CPM, PERT, resource leveling — each has its place

Apply the Principles

Sequence, estimate, allocate, monitor — consistently and deliberately

Stay Adaptable

The best schedulers treat disruption as expected, not exceptional

> **Across every industry and every endeavor, those who master their schedule master their outcomes.**